Statistics Questions – Step-by-Step Solutions & Answer Key

Question 1
Answer: b
Boxplots compare numerical distributions across categories.

Question 2
Answer: b
Mean=(1×6+2×11+3×9+4×3+5×1)/30=72/30=2.4≈2.3.

Question 3
Answer: a
Range=max–min=4–2=2.

Question 4
Answer: c
Explained by the Law of Large Numbers.

Question 5
Answer: a
z=(12–5)/2=3.5, more than 3 SD away.

Question 6
Answer: a
99% CI≈(60.5%,63.5%).

Question 7
Answer: Correct
CI = 3.2 ± 2.33 × (1.74/√50).

Question 8
Answer: a
t=0.5 gives p-value > 0.1.

Question 9
Answer: d
p=0.024<0.05, reject equal means.

Question 10
Answer: a
ANOVA compares means of several groups.

Question 11
Answer: b
75% of dependent variable variation explained.

Question 12
Answer: c
Age is continuous.

Question 13
Answer: d
Histogram bars have no gaps.

Question 14
Answer: d
Q1 = median of lower half.

Question 15
Answer: a
Both mean and SD multiply by 3.

Question 16
Answer: a
Range=2.

Question 17
Answer: e
Range is sensitive to outliers.

Question 18
Answer: g
Law of Large Numbers.

Question 19
Answer: c
Normal data gives roughly straight probability plot.

Question 20
Answer: b
Higher confidence → wider interval.

Question 21
Answer: c
Statements II and III are true.

Question 22
Answer: a
Small p-value means inconsistency with H0.

Question 23
Answer: b
$p=0.0294<0.10$ so reject H0.

Question 24
Answer: d
MBA managers have higher salaries.

Question 25
Answer: a
Within-group variation is random variation.

Question 26
Answer: d
ANOVA and t-test give same p-value for 2 groups.

Question 27
Answer: a
Large B compared to W supports factor effect.

Question 28
Answer: a
Eta squared ranges from 0 to 1.

Question 29
Answer: b
Same items compared → paired one-sided t-test.

Question 30
Answer: d
Difference below 70cm³, not clinically important.

Question 31
Answer: a
One-tailed used when direction is specified.

Question 32
Answer: b
Same result as Q23.

Question 33
Answer: c
Failing to reject false H0 = Type II error.

Question 34
Answer: a
95% CI closest to (80.3%,83.6%).

Question 35
Answer: c
Approx SD ≈ range/6 = 15.

Question 36
Answer: d
75th percentile z≈0.675.

Question 37
Answer: c
Sample SD≈1.67.

Question 38
Answer: IQR
Width of box = interquartile range.

Question 39
Answer: a
Mean=150 and SD=30.

Question 40
Answer: a
Q1 is median of lower half.

Question 41
Answer: c
Colors are qualitative.

Question 42
Answer: a
Population counts are discrete.

Question 43
Answer: c
Yes/no and rankings are qualitative.

Question 44
Answer: b
Distribution is right skewed.

Question 45
Answer: d
Final concentration=33.6%.

Question 46
Answer: c
Equal means are not an ANOVA assumption.

Question 47
Answer: d
Large differences favor HA.

Question 48
Answer: 3rd option
Correct formula: $3.2 \pm 2.33\times(1.74/\sqrt{50})$.

Question 49
Answer: 0.0688
Closest p-value choice.

Question 50
Answer: a
Sample space = all possible outcomes.

Question 51
Answer: c
$1-(1/2)^3=7/8$.

Question 52
Answer: Yes
$p<0.001$, reject equal means.

Question 53
Answer: d
p-value between 0.02 and 0.05.

Question 54
Answer: b
1200 is inside confidence interval.

Question 55
Answer: a
Temperature is continuous.

Question 56
Answer: b
Smallest spread.

Question 57
Answer: e
Mutually exclusive events cannot be independent.

Question 58
Answer: c
Approx SD=15.

Question 59
Answer: a
All statements are true.

Question 60
Answer: d
Binomial not for counting animals in one day.

Question 61
Answer: d
Two-sided alternative hypothesis.

Question 62
Answer: c
Rejecting true H0 = Type I error.